

Progress Report

A Unified Benchmark of Grokking Predictors in Neural Networks

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1. Objectives Completed

- Grokking reproduction gate: canonical grokking behaviour reproduced on $(a + b) \bmod 97$, completed July 28, 2026 (on schedule).
- Main benchmark protocol (Nanda-Unified v4) established: 4-head transformer, $(a + b) \bmod 113$, 5 seeds, 40,000 epochs — completed and reproducible from the committed config.
- Predictor 1, L2 Norm: implemented, benchmarked across 5 seeds, formally closed.
- Predictor 2, Dropout Variance: implemented, benchmarked across 5 seeds, formally closed.
- Predictor 3, Spectral (task-model alignment): implemented, benchmarked across 5 seeds, formally closed.
- Predictor 4, AGE (Neural Collapse, NC1): implemented, benchmarked across 5 seeds, formally closed.

Not yet completed: Predictors 5 through 9 (HTSR Alpha, Correlation Traps, Weight-PCA, Higher-order Mutual Information, Commutator Defect). Predictor 5 has not been started.

2. Work Accomplished

The reproduction gate was cleared first, as required before any predictor benchmarking. A single-head transformer trained on $(a + b) \bmod 97$ for 10,000 epochs showed the standard grokking signature: training accuracy reached 1.0000 by roughly epoch 935, test accuracy crossed 0.50 around epoch 3406 and 0.90 around epoch 4271, and finished at 0.9968 test accuracy by epoch 10,000.

The main benchmark then reproduced the architecture and training protocol of Nanda et al. (2023), “Progress Measures for Grokking via Mechanistic Interpretability” (ICLR 2023): 4-head transformer, model dimension 128, no LayerNorm, no linear biases, AdamW with betas (0.9, 0.98) and weight decay 1.0, small-init scheme (standard deviation approximately 0.0707), full-batch training on $(a + b) \bmod 113$ for 40,000 epochs, with 24 log-uniform checkpoints across 5 independent seeds. All 5 seeds grokked, with no non-convergent runs.

Against this protocol, four predictors were implemented, faithfully to their source papers, and evaluated with a single pre-registered falsification criterion each: does the predictor’s signal reach its extremum before the grok epoch, consistently across all 5 seeds. Each of the four was tested and formally closed. A secondary architectural variant, adding LayerNorm under the same weight-decay regime, was also tested; it destabilised training and was not adopted, and is retained in the repository as a documented negative result rather than discarded.

3. Challenges Encountered

- Large seed-to-seed variance in the grok epoch itself (14474, 6988, 10418, 10193, 24021 across the 5 seeds; mean 13218.8, standard deviation 5900.6). This means any candidate predictor is being asked to

track a moving target rather than a fixed epoch, which made the falsification criterion harder to design carefully.

- The Dropout predictor investigation was closed and then reopened within the same week, after further literature search surfaced the Salah & Yevick dropout-variance formulation (arXiv:2507.11645), which gave a more faithful implementation than the first attempt.
- The first implementation of the Spectral predictor used a plain SVD-based approach and had to be discarded and rebuilt from scratch as a faithful implementation of Canatar, Bordelon & Pehlevan (2021) before it could be evaluated properly.
- Schedule slippage: the Gantt milestone for this reporting period (“all nine predictors verified”, targeted for September Week 4) will not be met on time. Four of nine are closed, and the fifth has not been started yet.

4. Results Obtained

All four completed predictors were closed with a negative result — a genuine falsification of each hypothesis, not an inconclusive finding, since each was implemented faithfully before being tested.

#	Predictor	Source	Observed behaviour	Verdict
1	L2 Norm	Nanda et al. 2023 (Fig. 7)	Moving-average crossover fires very early (epochs 103–121 across seeds), thousands of epochs before grok in 5/5 runs.	Closed — negative
2	Dropout (variance)	Salah & Yevick, arXiv:2507.11645	Peak epoch lands at or after the grok epoch in 5/5 runs (peak/grok ratio 1.05–3.92).	Closed — negative
3	Spectral (task–model alignment)	Canatar, Bordelon & Pehlevan 2021	No pre-grok spectral concentration observed; representation entropy stayed flat (8.24 → 7.94); alignment peak follows grok in 5/5 runs.	Closed — negative
4	AGE (Neural Collapse, NC1)	Papayan et al. 2020 (PNAS)	NC1 does collapse sharply (about 660-fold, from roughly 45 to 0.05) in all 5 seeds, but this collapse lags the grok epoch (ratio 1.67–3.92) rather than leading it.	Closed — negative
5	HTSR Alpha	Martin & Mahoney (weightwatcher)	Not yet started.	Pending
6	Correlation Traps	—	Not yet started.	Pending
7	Weight-PCA	—	Not yet started.	Pending
8	Higher-order Mutual Information	—	Not yet started.	Pending
9	Commutator Defect	—	Not yet started.	Pending

Table 1. Status of all nine predictors under investigation. Rows 1–4 report completed, falsified results from the Nanda-Unified v4 benchmark (git commit 680a108, tag benchmark-v4-4predictors, September 7, 2026). Rows 5–9 are not yet started.

Read together, the pattern across the four closed predictors is itself informative: L2 Norm fails in the opposite direction from the other three (it fires far too early, while Dropout-variance, Spectral alignment, and AGE all fire at or after grok). No single one of these four signals, on its own, gives an early warning of generalisation on this task.

5. Plans for the Next Reporting Period

- Implement Predictor 5, HTSR Alpha (Martin & Mahoney, per-weight-matrix heavy-tailed self-regularisation exponent), using the weightwatcher library already listed as a project dependency.
- Continue in the established evaluation order through Predictors 6–9 (Correlation Traps, Weight-PCA, Higher-order Mutual Information, Commutator Defect), following the same implement → test → close cadence used for Predictors 1–4.
- Reorganise the thesis repository into the directory structure requested (01_Admin through Archive), and begin filing weekly progress reports and meeting minutes under 02_Progress_Reports and 03_Meeting_Minutes respectively, going forward.
- Revisit the remaining Gantt schedule with the supervisor, given the September Week 4 milestone will not be met, and confirm whether the October (leaderboard) and November (ensemble, submission) milestones remain realistic.

Source data: results/nanda_unified/aggregate.json and results/nanda_unified/BENCHMARK-V4.md (git commit 680a108);
Phase-1 gate log: docs/table_raw.txt; project memory: context.md.